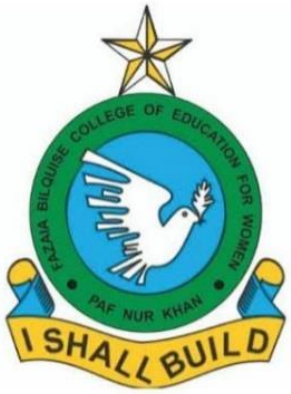
Fazaia Bilquis College of Education for Women PAF Base Nur Khan Base, Rawalpindi



Assignment 1

CLO-1: Implement and analyze data structures to solve simple applications (C3, GA3)

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Semester: 3rdB

Submitted to: Ms. Rabia Nasir

Course Name: Data Structures

Course Code: CS 216L

Submitted Date: 11 October 2026

STUDENT RECORD SYSTEM

Description:

The Student Record System is a menu-driven Java console application used to manage student records. It allows users to add, display, search, delete, update, and sort student information. It also allows users to undo the last delete or update operation.

Data Structures used:

Singly Linked List: Stores student records.

Stack: Stores previous student details for undo operations.

Algorithms Used

Linear Search: Searches for a student by roll number.

Bubble Sort: Sorts student records by GPA from highest to lowest.

Features:

1. Add a student record.
2. Display all records.
3. Search by roll number.
4. Delete a student record.
5. Update student details.
6. Sort by GPA.
7. Undo the last delete or update.
8. Exit the program.

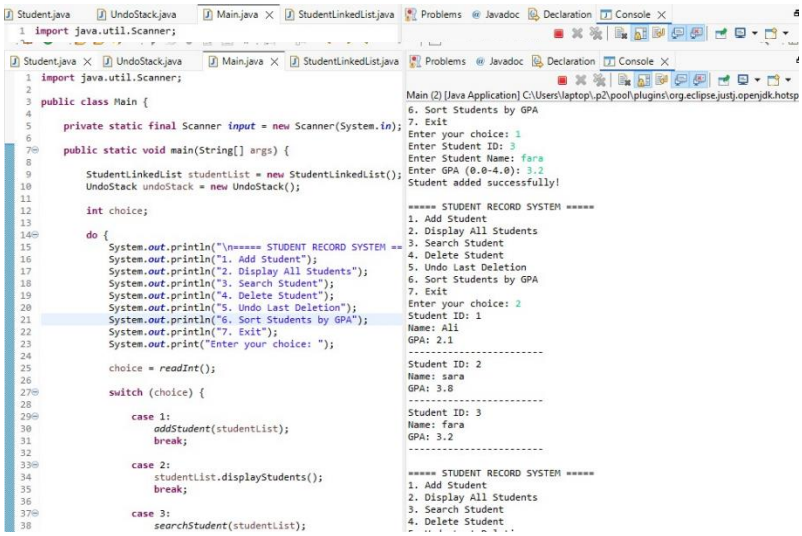
Time Complexity:

Operation	Time Complexity	Explanation
Insert Student	O(n)	Traverses the list to add a student at the end.
Delete Student	O(n)	Searches for the student and removes the node.
Search Student	O(n)	Checks records one by one using linear search.
Sort Students	$O(n^2)$	Bubble Sort compares student GPAs repeatedly.
Stack Push	O(1)	Adds an action to the top of the stack.
Stack Pop	O(1)	Removes an action from the top of the stack.
Display Students	O(n)	Visits every student record.
Update Student	O(n)	Searches for the student before updating details.

How to Run the Program:

1. Install the Java Development Kit (JDK).
2. Open the project in Eclipse.
3. Create or open StudentRecordSystem.java and paste the provided source code.
4. Run the class. The menu will appear in the Console.

Screenshots:



```
import java.util.Scanner;

public class Main {

    private static final Scanner input = new Scanner(System.in);

    public static void main(String[] args) {

        StudentLinkedList studentList = new StudentLinkedList();
        UndoStack undoStack = new UndoStack();

        int choice;

        do {
            System.out.println("\n===== STUDENT RECORD SYSTEM ==
            System.out.println("1. Add Student");
            System.out.println("2. Display All Students");
            System.out.println("3. Search Student");
            System.out.println("4. Delete Student");
            System.out.println("5. Undo Last Deletion");
            System.out.println("6. Sort Students by GPA");
            System.out.println("7. Exit");
            System.out.print("Enter your choice: ");

            choice = readInt();

            switch (choice) {

                case 1:
                    addStudent(studentList);
                    break;

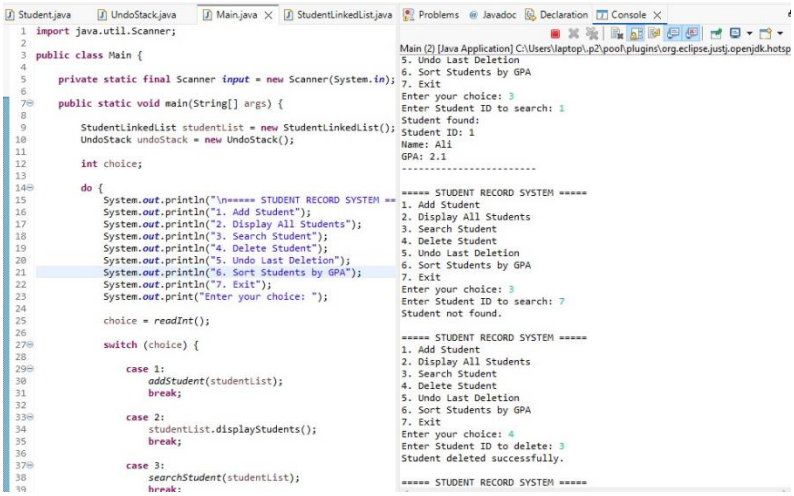
                case 2:
                    studentList.displayStudents();
                    break;

                case 3:
                    searchStudent(studentList);
```

```
Main (2) [Java Application] C:\Users\laptop\.p2\pool\plugins\org.eclipse.justj.openjdk.hotsp
6. Sort Students by GPA
7. Exit
Enter your choice: 1
Enter Student ID: 3
Enter Student Name: fara
Enter GPA (0.0-4.0): 3.2
Student added successfully!

===== STUDENT RECORD SYSTEM =====
1. Add Student
2. Display All Students
3. Search Student
4. Delete Student
5. Undo Last Deletion
6. Sort Students by GPA
7. Exit
Enter your choice: 2
Student ID: 1
Name: Ali
GPA: 2.1
-----------------------
Student ID: 2
Name: sara
GPA: 3.8
-----------------------
Student ID: 3
Name: fara
GPA: 3.2
-----------------------

===== STUDENT RECORD SYSTEM =====
1. Add Student
2. Display All Students
3. Search Student
4. Delete Student
```



```
import java.util.Scanner;

public class Main {

    private static final Scanner input = new Scanner(System.in);

    public static void main(String[] args) {

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        do {
            System.out.println("\n===== STUDENT RECORD SYSTEM ==
            System.out.println("1. Add Student");
            System.out.println("2. Display All Students");
            System.out.println("3. Search Student");
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            System.out.println("7. Exit");
            System.out.print("Enter your choice: ");

            choice = readInt();

            switch (choice) {

                case 1:
                    addStudent(studentList);
                    break;

                case 2:
                    studentList.displayStudents();
                    break;

                case 3:
                    searchStudent(studentList);
                    break;
```

```
Main (2) [Java Application] C:\Users\laptop\.p2\pool\plugins\org.eclipse.justj.openjdk.hotsp
5. Undo Last Deletion
6. Sort Students by GPA
7. Exit
Enter your choice: 3
Enter Student ID to search: 1
Student found:
Student ID: 1
Name: Ali
GPA: 2.1
-----------------------

===== STUDENT RECORD SYSTEM =====
1. Add Student
2. Display All Students
3. Search Student
4. Delete Student
5. Undo Last Deletion
6. Sort Students by GPA
7. Exit
Enter your choice: 3
Enter Student ID to search: 7
Student not found.

===== STUDENT RECORD SYSTEM =====
1. Add Student
2. Display All Students
3. Search Student
4. Delete Student
5. Undo Last Deletion
6. Sort Students by GPA
7. Exit
Enter your choice: 4
Enter Student ID to delete: 3
Student deleted successfully.

===== STUDENT RECORD SYSTEM =====
```

GitHub Repository Link:

GitHub URL: https://github.com/bscs14f2521-bot/CS216L_Project1_ansa.git